



2010
Preliminary Course
FINAL EXAMINATION

Mathematics

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using a black or blue pen
- Approved calculators may be used.
- All necessary working should be shown for every question.
- Begin each question on a fresh sheet of paper.

Total Marks (84)

- Attempt Questions 1 – 7
- All questions are of equal value.

Question 1: (12 marks) *Use a separate sheet of paper***Marks**

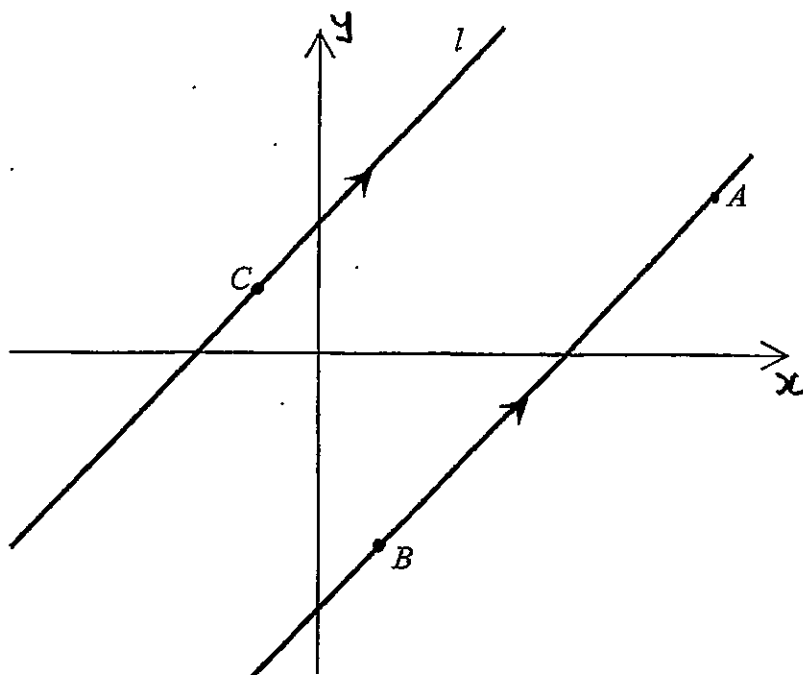
- (a) Express the decimal $0.\dot{2}0\dot{3}$ as a fraction in simplest form. 1
- (b) Calculate $\frac{3}{2.78} + \frac{1}{\sqrt{3}}$ correct to 2 decimal places. 1
- (c) If $\sqrt{27} + \sqrt{12} = \sqrt{A}$, find A 2
- (d) Expand and simplify $(5x - 2y)^2$ 1
- (e) Simplify $\frac{x^2-9}{x^2+x-12}$ 2
- (f) Factorise fully: $4x^3 - 32$ 2
- (g) Solve simultaneously 2
 $y = 3x - 7$
and $2x + y + 2 = 0$
- (h) Solve for x: 1
 $\frac{3x}{2} + 3 = x - 4$

Question 2: (12 marks) *Use a separate sheet of paper***Marks**

- (a) Evaluate $\sqrt[3]{6.91 \times 10^{-5}}$ correct to three significant figures. 2
- (b) Factorize $2x^2 + x - 28$. 2
- (c) Simplify $\frac{2x+3}{3} - \frac{x+2}{4}$. 2
- (d) Express $(2\sqrt{3}+1)(2-\sqrt{3})$ in the form $a\sqrt{3}+b$. 1
- (e) A pair of jeans were discounted by 15% to a selling price of \$63.75. Find the original marked price of the jeans before the discount was applied. 2
- (f) Solve the following inequality. $|3x-5| \geq 2$. 3

Question 3: (12 marks) *Use a separate sheet of paper*

(a)



NOT TO SCALE

The line l passes through $C(-1, 2)$ and has equation $y = 2x + 4$.

The point B has coordinates $(1, -6)$ and the line AB is parallel to line l .

Marks

- | | | |
|--------|---|---|
| (i) | Copy the above diagram, writing the coordinates of B and C onto this diagram. | 1 |
| (ii) | Find the length of the interval BC . | 1 |
| (iii) | Find the midpoint of BC . | 1 |
| (iv) | Write down the slope of the line l and find the angle l makes with the positive x -axis. (answer to the nearest degree) | 2 |
| (v) | Show that AB has equation $y = 2x - 8$. | 1 |
| (vi) | If P is a point which lies on AB and on the line $y = 2$, find the coordinates of P . | 1 |
| (vii) | Show that the perpendicular distance of P from the line l is $\frac{12\sqrt{5}}{5}$ units | 2 |
| (viii) | Find the size of $\angle ABC$ to the nearest minute. | 1 |
| (ix) | Find the exact area of triangle PBC | 2 |

Question 4: (12 marks) *Use a separate sheet of paper***Marks**

- (a) Find in simplest form $\frac{f(x)-f(c)}{x-c}$ if $f(x) = 5x + 7$ 2

- (b) Determine whether the function $f(x) = \frac{1}{x^2 - 4}$ is odd, even or neither odd nor even. WORKING MUST BE SHOWN. 1

- (c) Consider $f(x) = \begin{cases} -3 & \text{if } x \leq -1 \\ 2x - 3 & \text{if } x > -1 \end{cases}$. Evaluate $f(-1) + f(1)$. 1

- (d) Find the range of the function 1

$$y = \sqrt{4 - x^2}$$

- (e) **Sketch** graphs of the following functions and state the **domain** of each.

(i) $y = \frac{3}{2x - 1}$. 2

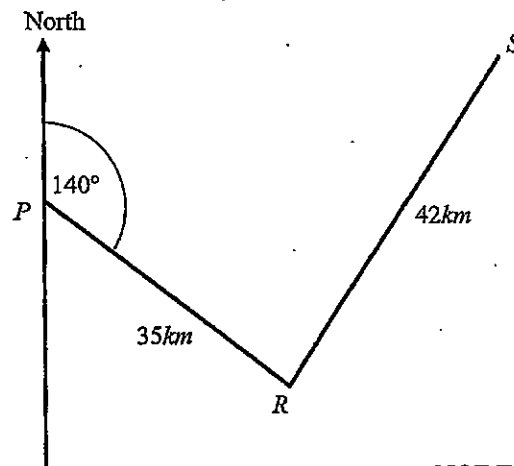
(ii) $y = |2 - 3x|$. 2

- (f) Show the region of the number plane where the following hold simultaneously:

$$\begin{aligned} (x-1)^2 + y^2 &\leq 9 \\ y &< x + 1 \end{aligned}$$
 3

Question 5: (12 marks) *Use a separate sheet of paper***Marks**(a) Find the exact value of $\cot 330^\circ$ 1(b) Solve $2\sin^2 x = 1$ where $-180^\circ \leq x \leq 180^\circ$ 2

(c)



NOT TO SCALE

A tourist drives 35km from the town of Pine Vale (P) on a bearing of 140° T to the town of Radiatagrove (R).

He then drives 42km on a bearing of 38° T to the town of Spruceville (S).

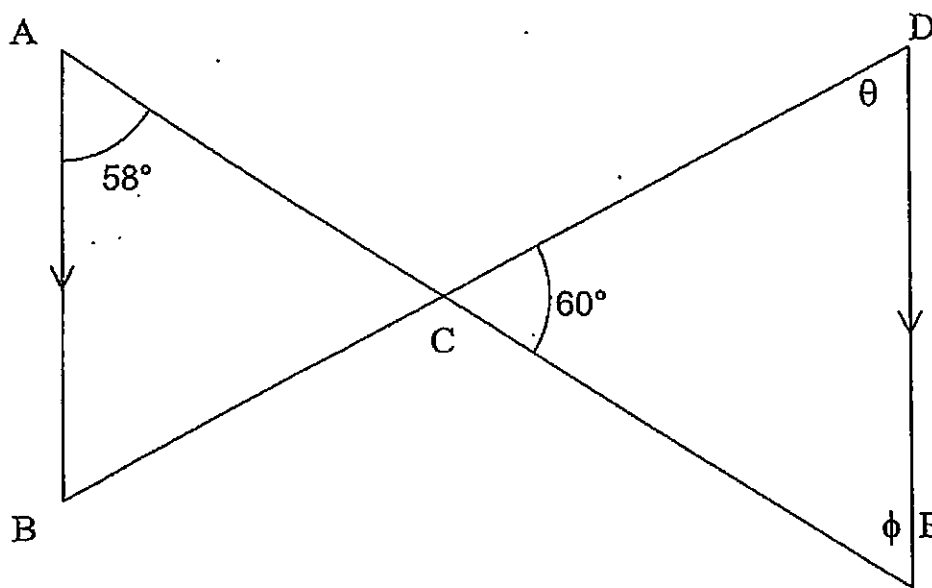
Copy this diagram(i) Show that $\angle PRS = 78^\circ$. 1(ii) Show that the distance from Spruceville to Pine Vale (SP) is 49km , correct to the nearest kilometre. 1(iii) Show the size of $\angle SPR = 57^\circ$ to the nearest degree. 1

(iv) Hence, or otherwise, find the bearing of Pine Vale from Spruceville. Show all necessary working. 2

(d) If $\sin \theta = -\frac{4}{11}$ and $\tan \theta > 0$ find the exact value of $\cos \theta$. 2(e) Prove that $\tan \theta + \cot \theta = \sec \theta \operatorname{cosec} \theta$ 2

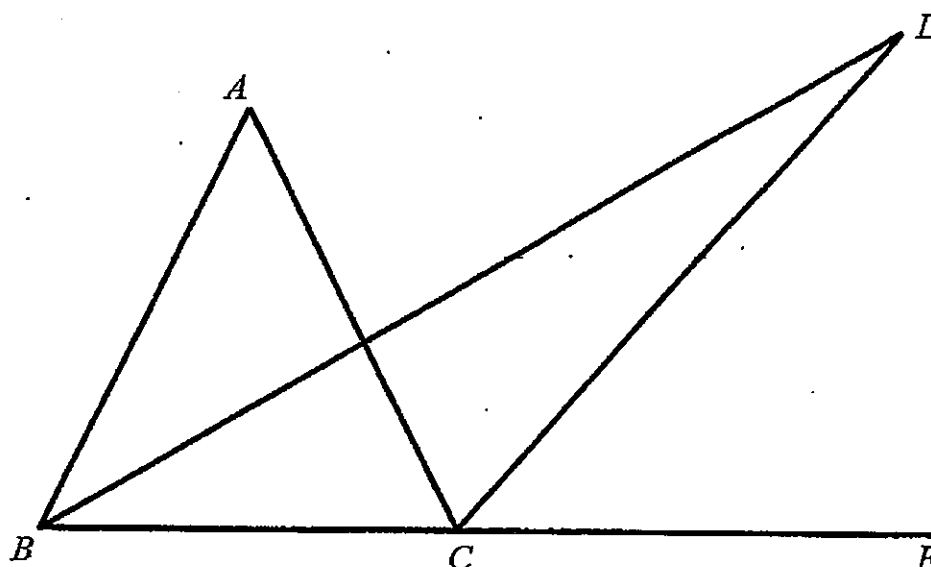
Question 6: (12 Marks) *Use a separate sheet of paper***Marks**

- (a) Find
- θ
- and
- ϕ



1

- (b)



NOT TO SCALE

ABC is an isosceles triangle in which $AB = AC$ and $\angle BAC = 64^\circ$.
 BC is produced to E . BD bisects $\angle ABC$ and CD bisects $\angle ACE$.

Copy or trace the diagram and mark on it all the given information.

- (i) Find the size of
- $\angle ABC$
- giving reasons.

1

- (ii) Find the size of
- $\angle BDC$
- giving reasons.

2

Question 6 (*continued*)**Marks**

- (c) The sum of the interior angles of a regular polygon is 3960° .

(i) How many sides has the polygon?

1

(ii) Find the size of each interior angle.

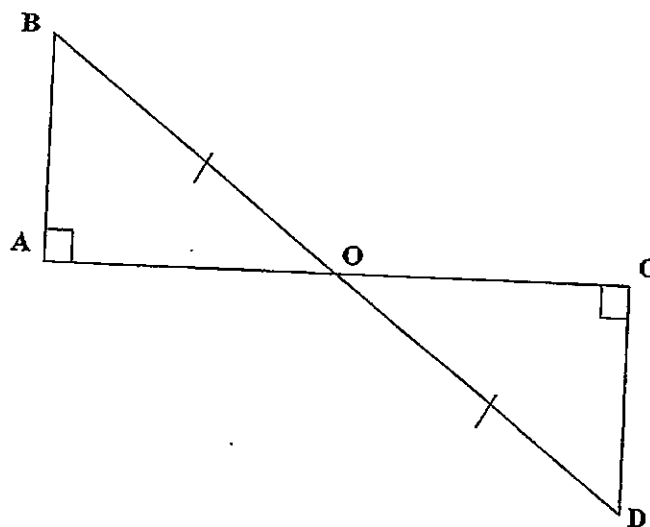
1

(iii) Hence or otherwise find the size of each exterior angle.

1

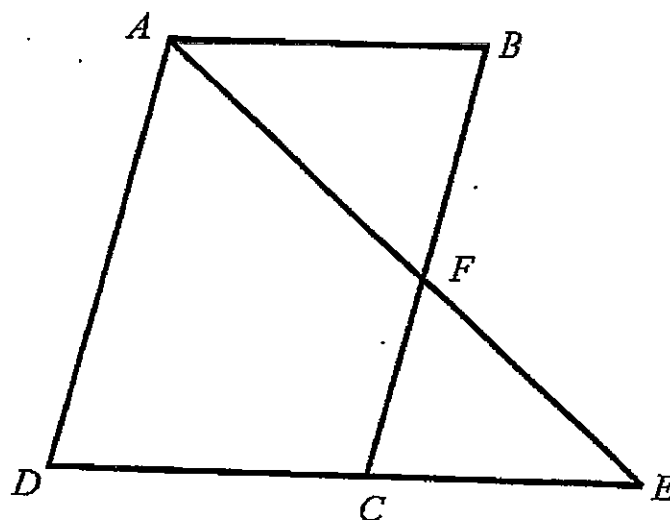
(d)

2



Prove that $\triangle AOB$ is congruent to $\triangle COD$

(e)



$ABCD$ is a parallelogram. DC is produced to E . AE cuts BC at F .
 $AD = 16\text{cm}$, $CE = 9\text{cm}$ and $BF = 10\text{cm}$.

- (i) Prove that $\triangle ABF$ is similar to $\triangle ECF$

2

- (ii) Find AB

1

Question 7: (12 marks) *Use a separate sheet of paper***Marks**

- (a) Express $9x^2 + 2x - 5$ in the form
 $ax(x + 1) + b(x + 1) + C$ 2
- (b) For what values of k will the expression $kx^2 - 4x + k$ always be positive? 2
- (c) Given the equation $3x^2 + 7x - 4 = 0$ has roots α and β ,
without finding α or β evaluate $\alpha^2 + \beta^2$ 2
- (d) Find the value of k for which the equation $x^2 - (k + 4)x + (k - 3) = 0$ has
(i) one root equal to -2
(ii) roots which are reciprocals of one another. 2
- (e) Find all real numbers x which satisfy the equation $x^4 = 8(x^2 + 6)$. 2
- (f) Find, as a relationship between a , b and c , the condition for the
quadratic equation in x
$$(a^2 - b^2)x^2 + 2b(a - c)x + (b^2 - c^2) = 0$$

to have equal roots. Simplify your answer as far as possible. 2

END OF EXAM

QUESTION 1

a) $x = 0.\dot{2}0\dot{3}$

$1000x = 203.\dot{2}0\dot{3}$

$\therefore 999x = 203$ ①

$x = \frac{203}{999}$

b) $\frac{3}{2.78} + \frac{1}{\sqrt{3}} = 1.65648696\dots$
(calc) ①
 $= 1.66$ (2d.p.)

c) $\sqrt{27} + \sqrt{12} = 3\sqrt{3} + 2\sqrt{3}$
 $= 5\sqrt{3}$
 $= \sqrt{75}$ ②
 $\therefore A = 75$

d) $(5x - 2y)^2 = 25x^2 - 20xy + 4y^2$ ①

e) $\frac{x^2 - 9}{x^2 + x - 12} = \frac{(x-3)(x+3)}{(x+4)(x-3)}$ ②
 $= \frac{x+3}{x+4}$

f) $4x^3 - 32 = 4(x^3 - 8)$ ②
 $= 4(x-2)(x^2 + 2x + 4)$

g) $y = 3x - 7$ - (1)
 $2x + y + 2 = 0$ - (2)

SUB ① in ②

$2x + (3x - 7) + 2 = 0$

$5x - 5 = 0$

$5x = 5$

$x = 1$ ②

SUB in ① $y = -4$

$\therefore x = 1, y = -4$

h) $\frac{3x}{2} + 3 = x - 4$
 $\frac{x}{2} = -7$ ①
 $x = -14$

QUESTION 2

a) $\sqrt[3]{6.91 \times 10^{-5}} = 0.04103564$ (calc) ②
 $= 0.0410$ (3 s.f.)

b) $2x^2 + x - 28 = \frac{(2x+8)(2x-7)}{2}$ ②
 $= (x+4)(2x-7)$

c) $\frac{2x+3}{3} - \frac{x+2}{4}$
 $= \frac{4(2x+3) - 3(x+2)}{12}$
 $= \frac{8x+12 - 3x-6}{12}$ ②
 $= \frac{5x+6}{12}$

$$\begin{aligned} d) & (2\sqrt{3}+1)(2-\sqrt{3}) \\ &= 4\sqrt{3} - 6 + 2 - \sqrt{3} \\ &= 3\sqrt{3} - 4 \end{aligned}$$

①

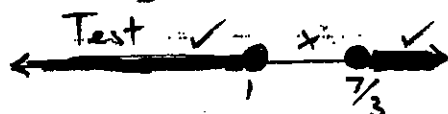
$$\begin{aligned} e) \quad 85\% &= \$63.75 \\ 1\% &= \$63.75 \div 85 = \$0.75 \\ 100\% &= \$0.75 \times 100 \\ &= \$75 \\ \therefore \text{Cost} &= \$75 \end{aligned}$$

②

$$\begin{aligned} f) \quad |3x-5| &\geq 2 \\ \text{Consider } |3x-5| &= 2 \end{aligned}$$

$$\begin{aligned} 3x-5 &= 2 \quad \text{or} \quad 3x-5 = -2 \\ 3x &= 7 \quad \quad \quad 3x &= 3 \\ x &= \frac{7}{3} \quad \quad \quad x &= 1 \end{aligned}$$

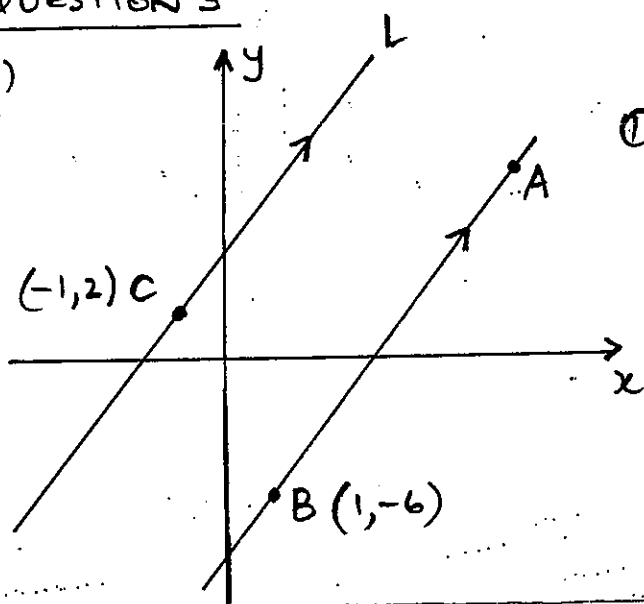
③



$$\therefore x \leq 1, x \geq \frac{7}{3}$$

QUESTION 3

(1)



①

$$\begin{aligned} \text{(ii)} \quad BC &= \sqrt{(1+1)^2 + (-6-2)^2} \\ &= \sqrt{4 + 64} \\ &= \sqrt{68} \\ &= 2\sqrt{17} \text{ units} \end{aligned}$$

①

$$\begin{aligned} \text{(iii)} \quad \text{Midpt}_{BC} &= \left(\frac{1+1}{2}, \frac{-6+2}{2} \right) \\ &= (0, -2) \end{aligned}$$

①

$$\begin{aligned} \text{(iv)} \quad \text{Since } L &\Rightarrow y = 2x + 4 \\ \therefore m &= 2 \\ \therefore \text{slope is } &2 \end{aligned}$$

②

$$\begin{aligned} \therefore \tan \theta &= 2 \\ \theta &= 63.4349\dots \\ \therefore \theta &= 63^\circ \text{ (nearest degree)} \end{aligned}$$

$$\begin{aligned} \text{(v)} \quad \text{Using } y - y_1 &= m(x - x_1) \\ \text{and since } AB &\parallel L, m = 2 \end{aligned}$$

$$\begin{aligned} y + 6 &= 2(x - 1) \\ y + 6 &= 2x - 2 \\ y &= 2x - 8 \\ &\text{as req.} \end{aligned}$$

①

$$\begin{aligned} \text{(vi)} \quad (y=2) \quad 2 &= 2x - 8 \\ 10 &= 2x \\ 5 &= x \end{aligned}$$

①

$$\therefore P = (5, 2)$$

$$\text{(vii)} \quad L \Rightarrow 2x - y + 4 = 0$$

$$\text{Id} = \frac{|2(5) - 1(2) + 4|}{\sqrt{2^2 + (-1)^2}}$$

$$= \frac{12}{\sqrt{5}} = \frac{12\sqrt{5}}{5} \text{ units as req.}$$

(viii) $\perp$ of C from AB = $\frac{12\sqrt{5}}{5}$

$$\therefore \sin \theta = \frac{\frac{12\sqrt{5}}{5}}{2\sqrt{17}}$$

$$= \frac{6\sqrt{5}}{5\sqrt{17}}$$

$$\therefore \theta = 40^\circ 36'$$

(ix) Area of $\triangle ABC = \frac{1}{2} AB \times \frac{12\sqrt{5}}{5}$

$$d_{AB} = \sqrt{(5-1)^2 + (2+6)^2}$$

$$= \sqrt{16 + 64}$$

$$= \sqrt{80}$$

$$= 4\sqrt{5}$$

$$\text{Area} = \frac{1}{2} \times 4\sqrt{5} \times \frac{12\sqrt{5}}{5}$$

$$= 24 \text{ units}^2$$

QUESTION 4

a) $\frac{f(x) - f(c)}{x - c} = \frac{5x + 7 - 5c - 7}{x - c}$

$$= \frac{5x - 5c}{x - c}$$

$$= \frac{5(x - c)}{x - c}$$

$$= 5$$

b) $f(x) = \frac{1}{x^2 - 4}$

$$f(-x) = \frac{1}{(-x)^2 - 4} = \frac{1}{x^2 - 4}$$

$\therefore$ even since $f(x) = f(-x)$

c) $f(x) = \begin{cases} -3 & x \leq -1 \\ 2x - 3 & x > -1 \end{cases}$

$$\therefore f(-1) = -3$$

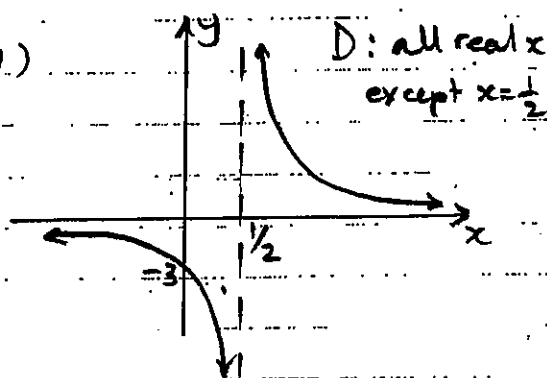
$$f(1) = 2(1) - 3 = -1$$

$$\therefore f(-1) + f(1) = -3 + (-1) = -4$$

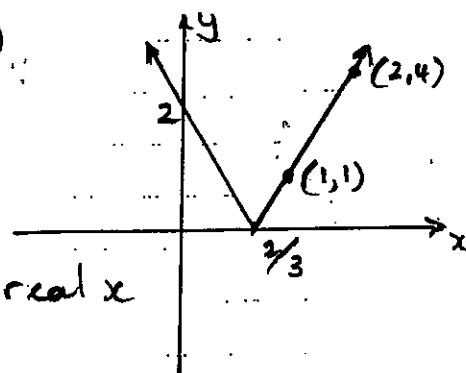
d) Semi-circle

$$\therefore \text{range} \Rightarrow 0 \leq y \leq 2$$

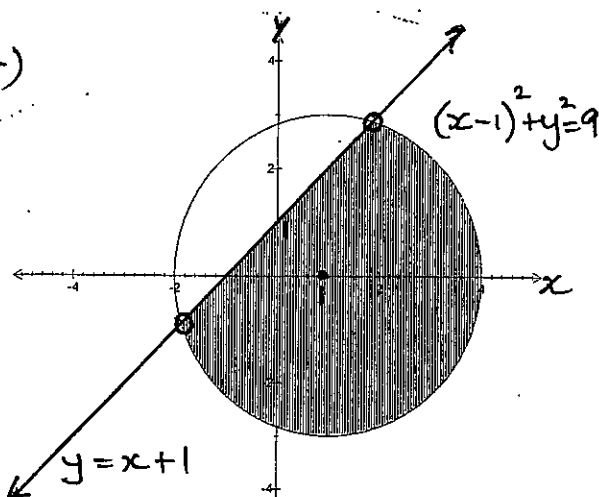
e) (i)



(ii)



f)

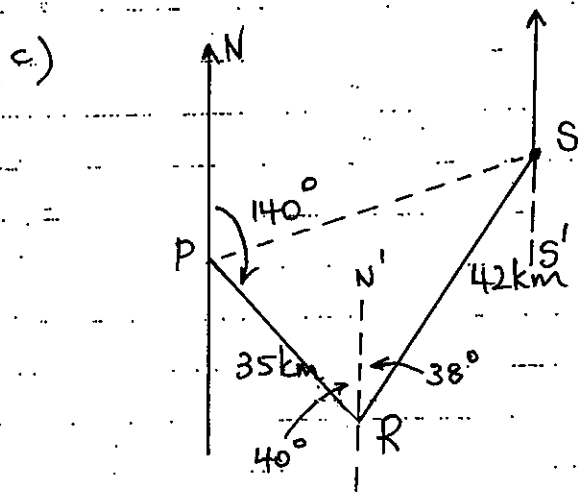


QUESTION 5

a) $\cot 330^\circ = -\cot 30^\circ$
 $= -\sqrt{3}$ ①

b) $2\sin^2 x = 1$
 $\sin^2 x = \frac{1}{2}$
 $\sin x = \pm \frac{1}{\sqrt{2}}$ ②

$\therefore x = -135^\circ, -45^\circ, 45^\circ, 135^\circ$



(i) $\hat{PRN}' = 40^\circ$ (co-interior $\angle$'s in parallel lines are supplementary)

$\hat{N'RS} = 38^\circ$ (given in data)

$\therefore \angle PRS = 40 + 38 = 78^\circ$ ①

(ii) $SP^2 = 35^2 + 42^2 - 2 \times 35 \times 42 \times \cos 78^\circ$
 $= 2377.739629...$ (calc)

$\therefore SP = 48.76207...$ (calc)

$\therefore SP = 49$ (nearest whole no.) ①

(iii) Let $\angle SPR = \theta$

$\frac{\sin \theta}{42} = \frac{\sin 78^\circ}{SP}$

$\sin \theta = \frac{42 \sin 78^\circ}{SP}$ ①

$= 0.8425...$ (calc)

$\therefore \theta = 57.4053...$ (calc)

$\therefore \angle SPR = 57^\circ$ (nearest degree)

(iv) $\angle PSR = 180 - (78 + 57)$ ($\angle$ sum of $\Delta = 180^\circ$)
 $= 45^\circ$

$\angle RSS' = 38^\circ$ (alternate $\angle$'s in parallel lines are equal) ②

$\therefore \text{Bearing} = 180 + 38 + 45$
 $= 253^\circ \text{ T.}$

d)

Using this triangle $\cos \theta = \frac{\sqrt{105}}{11}$

since $\sin \theta < 0$ $\tan \theta > 0$

$\therefore$ in 3rd Quad and $\cos \theta < 0$

$\therefore \cos \theta = -\frac{\sqrt{105}}{11}$

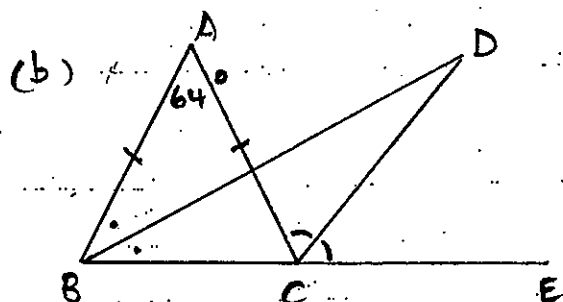
e) LHS = $\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$

$= \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta}$

$= \frac{1}{\cos \theta \sin \theta}$

$= \frac{1}{\cos \theta} \cdot \frac{1}{\sin \theta} = \sec \theta \csc \theta$
 $= \text{RHS}$

QUESTION 6 (a) $\phi = 58$, $\theta = 62$ ①



(i) Since $\triangle ABC$ isosceles

$$\therefore \angle ABC = \angle ACB$$

$$\therefore \angle ABC = 58^\circ \quad (\angle \text{sum of } \triangle = 180^\circ)$$

(ii) $\angle DBC = 29^\circ \quad (\frac{1}{2} \times \angle ABC)$

$$\angle BCD = 58^\circ + \frac{1}{2} \times \angle ACE$$

$$\angle ACE = 116^\circ \quad (\text{supplementary to } \angle ACB)$$

$$\therefore \angle BCD = 116^\circ$$

$$\text{and } \angle BDC = 35^\circ \quad (\angle \text{sum of } \triangle = 180^\circ)$$

(c)(i) $(n-2) \times 180 = 3960$

$$n-2 = 22$$

$$n = 24$$

$\therefore$ polygon has 24 sides

(ii) $3960 \div 24 = 165$

$\therefore$ Each angle is 165° ①

(iii) Either $180 - 165 = 15^\circ$

$$\text{or } 360 \div 24 = 15^\circ$$

$\therefore$ Each exterior $\angle = 15^\circ$ ①

(d) $\angle BAO = \angle DCO = 90^\circ$ (given)

$$\angle BOA = \angle DOC \quad (\text{vertically opposite } \angle \text{'s are equal})$$

$$BO = DO \quad (\text{given}) \quad ②$$

$$\therefore \triangle AOB \equiv \triangle COD \quad (\text{AAS})$$

(e)(i) $\angle AFB = \angle CFE$ (vertically opposite $\angle \text{'s are equal}$)

$$\angle BAF = \angle FEC \quad (\text{alternate } \angle \text{'s in parallel lines are equal})$$

(parallel lines since sides of parallelogram and sides produced) ②

(ii) $\frac{AB}{CE} = \frac{BF}{CF}$ (corresponding sides are in proportion)

$$\frac{AB}{9} = \frac{10}{6}$$

$$6AB = 90$$

$$AB = 15 \quad ①$$

QUESTION 7

a) $9x^2 + 2x - 5 \equiv ax(x+1) + b(x+1) + c$

$$\text{RHS} = ax^2 + ax + bx + b + c$$

$$= ax^2 + (a+b)x + (b+c)$$

$$\therefore a = 9, \quad a+b = 2, \quad b+c = -5$$

$$9+b = 2 \quad -7+c = -5$$

$$b = -7 \quad c = 2$$

$$9x^2 + 2x - 5 = 9x(x+1) - 7(x+1) + 2$$

(b) Quadratic is always positive when $a > 0$ and $\Delta < 0$

$$\Delta = b^2 - 4ac$$

$$= 16 - 4k^2$$

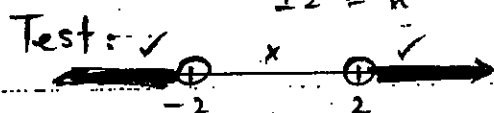
$$\therefore 16 - 4k^2 < 0$$

$$\text{Consider } 16 - 4k^2 = 0$$

$$16 = 4k^2$$

$$4 = k^2$$

$$\pm 2 = k$$



$$\therefore k < -2, k > 2$$

Since $a > 0$, $k > 2$ are only values.

$$(c) \quad 3x^2 + 7x - 4 = 0$$

$$\alpha + \beta = -\frac{7}{4}$$

$$\alpha\beta = -\frac{4}{3}$$

$$\text{Since } \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= \left(-\frac{7}{4}\right)^2 - 2\left(-\frac{4}{3}\right)$$

$$= \frac{49}{16} + \frac{8}{3}$$

$$= \frac{275}{48} \left(5 \frac{35}{48}\right)$$

$$(d) \quad x^2 - (k+4)x + (k-3) = 0$$

$$(x = -2) \quad 4 + (k+4)2 + (k-3) = 0$$

$$4 + 2k + 8 + k - 3 = 0$$

$$3k = -9$$

$$\therefore k = -3$$

(ii) let α, β be roots

$$\text{and since } \beta = \frac{1}{\alpha}$$

$$\text{product of roots } \Rightarrow \alpha \times \frac{1}{\alpha} = \frac{c}{a}$$

$$1 = k - 3$$

$$\therefore 4 = k$$

$$(e) \quad x^4 = 8(x^2 + 6)$$

$$x^4 = 8x^2 + 48$$

$$x^4 - 8x^2 - 48 = 0$$

$$(x^2 - 12)(x^2 + 4) = 0$$

$$\therefore x^2 = 12, -4$$

$$x = \pm\sqrt{12},$$

$$(x^2 = -4 \text{ no real solutions})$$

$$\therefore x = \pm 2\sqrt{3}$$

$$(f) \quad (a^2 - b^2)x^2 + 2b(a-c)x + (b^2 - c^2) = 0$$

For equal roots $\Delta = 0$

$$\Delta = [2b(a-c)]^2 - 4(a^2 - b^2)(b^2 - c^2)$$

$$= 4b^2(a-c)^2 - 4(a^2 - b^2)(b^2 - c^2)$$

$$= 4b^2(a^2 - 2ac + c^2) - 4(a^2b^2 - a^2c^2 - b^4 + b^2c^2)$$

$$= 4a^2b^2 - 8ab^2c + 4b^2c^2 - 4a^2b^2 + 4a^2c^2 + 4b^4 - 4b^2c^2$$

$$= 4b^4 + 4a^2c^2 - 8ab^2c$$

$$= 4(b^4 - 2ab^2c + a^2c^2)$$

$$= 4(b^2 - ac)^2$$

$$\therefore \text{For equal roots } b^2 - ac = 0$$

$$b^2 = ac$$